PROJECT TITLE

2D RUNNING GAME

PRESENTED BY

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Date of Submission

3/6/2026

Week 1: Project Planning and Requirement Analysis

- Analyze project requirements provided by the supervisor.
- Identify the main features of the game such as player movement, obstacles, score system, game over screen, restart option, and exit option.
- Prepare a basic workflow and development plan.
- Install and configure Java Development Kit (JDK) and development environment.

Week 2: Learning Java Swing and Designing the User Interface

- Study the basic components of Java Swing.
- Learn how to create JFrame, JPanel, and GUI components.
- Design the main game window.
- Create the initial project structure.

Week 3: Player Character Development

- Design the player character.
- Set the player's initial position on the screen.
- Implement player movement controls.
- Test character movement functionality.

Week 4: Obstacle Development

- Create obstacle objects and their properties.
- Implement obstacle generation logic.
- Add continuous obstacle movement from right to left.
- Test obstacle appearance and movement.

Week 5: Game Loop Implementation

- Develop the game loop using Java Swing Timer.
- Ensure continuous screen updates.
- Synchronize player and obstacle movement.
- Improve game responsiveness.

Week 6: Collision Detection and Game Over Logic

- Implement collision detection between player and obstacles.
- Stop the game when a collision occurs.
- Display a game over message.
- Test collision scenarios and fix issues.

Week 7: Score System Development

- Create a score calculation mechanism.
- Increase score based on survival time.
- Display the score during gameplay.
- Verify score accuracy.

Week 8: Play Again and Exit Functionality

- Design the game over screen.
- Add Play Again button to restart the game.
- Add Exit button to close the application.
- Test all game control options.

Week 9: Testing and Debugging

- Perform complete game testing.
- Identify and fix bugs.
- Improve user experience and gameplay.
- Optimize game performance.

Week 10: Documentation and Final Submission

- Prepare project documentation.
- Update GitHub repository and README file.
- Review all implemented features.
- Conduct final testing and prepare for project presentation and submission.